

Week 3: Seller and Inventory Management System

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Scenario:

The phone case business sells different types of phone cases supplied by multiple sellers. The system stores seller details, tracks the products supplied by each seller, and monitors stock levels to ensure phone cases are available.

1. Create Seller and Inventory Tables

```
SQL> CREATE TABLE Seller(
  2  Seller_ID VARCHAR(7),
  3  Seller_Name VARCHAR(30),
  4  Email VARCHAR(30),
  5  Phone NUMBER(10),
  6  Rating NUMBER(2,1)
  7 );

Table created.

SQL> CREATE TABLE Inventory(Inventory_ID VARCHAR(7),
  2  Product_ID VARCHAR2(7),
  3  Seller_ID VARCHAR(7),
  4  Stock_Quantity number,
  5  Stock_status VARCHAR(11),
  6  Last_Updated DATE);

Table created.
```

2. Establish Relationships

Seller and Product

- One seller can supply many phone cases.
- A phone case can be supplied by one or more sellers.

Relationship:

Seller → Product = One to Many

Product and Inventory

- Each phone case has stock information.
- Inventory tracks the available stock of each phone case.

Relationship:

Product → Inventory = One to One

Primary Key added for Both Tables:

```
SQL> ALTER TABLE Seller
  2  ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);

Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);

Table altered.

SQL> |
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);  
Table altered.
```

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT FK_Seller FOREIGN KEY(Seller_ID)  
3 REFERENCES Seller(Seller_ID);  
Table altered.
```

Unique Email ID:

```
SQL> ALTER TABLE Seller ADD  
2 CONSTRAINT Un_mail UNIQUE(Email);  
Table altered.
```

Email NOT NULL:

```
SQL> ALTER TABLE Seller MODIFY Email NOT NULL;  
Table altered.
```

Constraint to Check the Stock Quantity is Positive:

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT CK_Quan CHECK(Stock_Quantity >=0);  
Table altered.
```

Inserting Values to Seller Table:

```
SQL> INSERT INTO Seller VALUES('S000002','StyleCase Hub','stylecase@gmail.com',9876500002,4.2);
1 row created.

SQL>
SQL> INSERT INTO Seller VALUES('S000003','CoverWorld','coverworld@gmail.com',9876500003,4.7);
1 row created.

SQL>
SQL> INSERT INTO Seller VALUES('S000004','PhoneShells','phoneshells@gmail.com',9876500004,4.0);
1 row created.

SQL>
SQL> INSERT INTO Seller VALUES('S000005','CaseZone','casezone@gmail.com',9876500005,4.4);
1 row created.

SQL>
SQL> INSERT INTO Seller VALUES('S000006','CustomCovers','customcovers@gmail.com',9876500006,4.8);
1 row created.
```

```
SQL> SELECT * FROM Seller;
```

SELLER_	SELLER_NAME	EMAIL	PHONE
	RATING		
S000002	StyleCase Hub 4.2	stylecase@gmail.com	9876500002
S000003	CoverWorld 4.7	coverworld@gmail.com	9876500003
S000004	PhoneShells 4	phoneshells@gmail.com	9876500004
SELLER_	SELLER_NAME	EMAIL	PHONE
	RATING		
S000005	CaseZone 4.4	casezone@gmail.com	9876500005
S000006	CustomCovers 4.8	customcovers@gmail.com	9876500006

Inserting Values to Inventory Table

```
SQL> INSERT INTO Inventory VALUES('Inv0001','P000001','S000001',40,'Available',TO_DATE('2026-09-12','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv0002','P000002','S000002',35,'Available',TO_DATE('2026-09-13','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv0003','P000003','S000003',30,'Available',TO_DATE('2026-09-15','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv0004','P000004','S000004',25,'Available',TO_DATE('2026-09-14','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory
  2 VALUES('Inv0005','P000005','S000005',35,'Available',TO_DATE('2026-09-16','YYYY-MM-DD'));
1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES('Inv0006','P000006','S000006',20,'Available',TO_DATE('2026-09-17','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory
  2 VALUES('Inv0007','P000007','S000001',0,'Unavailable',TO_DATE('2026-09-17','YYYY-MM-DD'));
1 row created.
```

```
SQL> SELECT * FROM Inventory;
```

INVENTO	PRODUCT	SELLER_	STOCK_QUANTITY	STOCK_STATU	LAST_UPDA
Inv0001	P000001	S000001	40	Available	12-SEP-26
Inv0002	P000002	S000002	35	Available	13-SEP-26
Inv0003	P000003	S000003	30	Available	15-SEP-26
Inv0004	P000004	S000004	25	Available	14-SEP-26
Inv0005	P000005	S000005	35	Available	16-SEP-26
Inv0006	P000006	S000006	20	Available	17-SEP-26
Inv0007	P000007	S000001	0	Unavailable	17-SEP-26

7 rows selected.

Report 1: Seller-wise Product Report

```
SQL> SELECT s.Seller_Name,
  2         COUNT(i.Product_ID) AS Number_of_Products,
  3         SUM(i.Stock_Quantity) AS Available_Stock
  4 FROM Seller s
  5 JOIN Inventory i ON s.Seller_ID = i.Seller_ID
  6 WHERE i.Stock_Status = 'Available'
  7 GROUP BY s.Seller_Name;
```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
StyleCase Hub	1	35
CustomCovers	1	20
CaseCraft Store	1	40
CoverWorld	1	30
PhoneShells	1	25
CaseZone	1	35

6 rows selected.

Report 2: Stock Availability Report

```
SQL> SELECT p.Product_Name,  
2         i.Stock_Quantity,  
3         i.Stock_Status  
4 FROM Product p  
5 JOIN Inventory i ON p.Product_ID = i.Product_ID;
```

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATU

Clear Shock Case
40 Available

Soft Silicone Case
35 Available

Crystal Clear Case
30 Available

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATU

Floral Printed Case
25 Available

Luxury Designer Case
35 Available

Personal Photo Case
20 Available

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATU

Heavy Duty Case
0 Unavailable

Report 3: Inventory Analysis

Total products available:

```
SQL> SELECT COUNT(Product_ID) AS Total_Products_Available  
2 FROM Inventory  
3 WHERE Stock_Status = 'Available';
```

TOTAL_PRODUCTS_AVAILABLE

6

Products out of stock:

```
SQL> SELECT COUNT(Product_ID) AS Products_Out_of_Stock
  2   FROM Inventory
  3   WHERE Stock_Quantity = 0;

PRODUCTS_OUT_OF_STOCK
-----
                      1
```

Highest stocked products:

```
SQL> SELECT p.Product_Name,
  2         i.Stock_Quantity
  3   FROM Product p
  4   JOIN Inventory i ON p.Product_ID = i.Product_ID
  5   WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);

PRODUCT_NAME
-----
STOCK_QUANTITY
-----
Clear Shock Case
                40
```

Average inventory quantity:

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory_Quantity
  2   FROM Inventory;

AVERAGE_INVENTORY_QUANTITY
-----
                26.4285714
```